SOC corrected Gaussian basis sets for electronic transport calculations in ANT.Gaussian.

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Introduction

This is very brief summary of the implementation discussed in Pakdel et al. [1]. There we showed that if a GTO basis set describes properly the band structure of a material in the absence of SOC, then adding SOC as a correction in a post self-consistent-field (SCF) step gives a good approximation to the actual bands when compared to reference methods which produce very high quality results such as `VASP` [2], `Wien2k` [3], `Quantum Espresso` [4] or `OpenMX` [5–8].

This version (2026) recalibrates the SOC factors of every pseudopotential basis set against self-consistent two-component `CRYSTAL23` bands computed with the same basis set (see Calibration). Table 1 collects the factors; Using the factors in `ANT.Gaussian` explains how to use them, and the following sections give, for each element, the crystal structure, the basis set and pseudopotential in Gaussian format, and the band structure obtained with the calibrated factors.

Summary

Table 1. SOC factors for `ANT.Gaussian`. Structure and lattice parameters (Å; a/c for A7) of the calibration, basis set, pseudopotential (SC-n: Stuttgart small-core with n core electrons; AE: all-electron), number of basis functions per atom, SOC factors, Yukawa cutoff r_c (bohr, keyword `RCUT`) and RMS / maximum deviation (meV) from the two-component bands between $E_F - 6$ and $E_F + 4$ eV. Element symbols link to the detailed sections.

El	Str.	Latt.	Basis set	ECP	AOs	P	D	F	r_c	RMS / max
Au	fcc	4.05	pob-TZVP-rev2 + $s(0.1)$	SC-60	36	201.3	43.2	12.6	2.0	10.3 / 49
Ag	fcc	4.075	pob-DZVP-2018	SC-28	26	49.1	8.9	0.1	2.0	3.4 / 18
Pt	fcc	3.91	pob-TZVP-rev2	SC-60	35	229.4	41.6	9.3	2.0	5.0 / 24
W	bcc	3.16	pob-TZVP-rev2 (mod.)	SC-60	36	241.0	55.4	10.7	2.0	5.8 / 28
Pb	fcc	4.93	Baranek 2013	SC-60	33	248.3	66.5	0.0	2.0	9.9 / 52
Bi	A7	4.546/11.862	pob-TZVP-rev2 + $s(0.1)$	SC-60	37	230.9	24.1	–	2.0	11.7 / 41
Sb	A7	4.3007/11.222	pob-TZVP-2018	SC-28	36	76.9	0.0	–	2.0	5.8 / 24
Al	fcc	4.05	pob-DZVP-rev2	AE	18	1.0	1.0	0.0	2.5	–
Cu	fcc	3.63	Ruiz 2003	AE	39	1.0	1.0	0.0	2.5	–

Details

The Dirac-Kohn-Sham Hamiltonian can be written, to lowest order, simply adding a standard atomic SOC contribution [1] to the SOC-free Kohn-Sham Hamiltonian:

$$\xi\left(r\right)\mathbf{L}\cdot\mathbf{S}.$$

When represented in an atomic orbital basis set, it becomes

$$\left[\xi_{\mathrm{ij}}\left\langle l_{\mathrm{i}};m_{\mathrm{i}};s|\mathbf{L}\cdot\mathbf{S}|l_{\mathrm{j}};m_{\mathrm{j}};s'\right\rangle\right], \tag{1}$$

where

$$\xi_{\mathrm{ij}}=\mathrm{SOCFAC}_{L}\,\frac{\hbar^{2}e^{2}}{8\pi\varepsilon_{0}m_{e}^{2}c^{2}}\int_{0}^{\infty}\left(-\frac{1}{r}\frac{\mathrm{d}V_{\mathrm{eff}}\left(r\right)}{\mathrm{d}r}\right)\frac{R_{\mathrm{i}}\left(r\right)R_{\mathrm{j}}\left(r\right)}{\sqrt{N_{\mathrm{i}}N_{\mathrm{j}}}}\,r^{2}\mathrm{d}r. \tag{2}$$

In Eq. (2), $V_{\mathrm{eff}}\left(r\right)$ is the effective nuclear potential [1] in units of $e/4\pi\varepsilon_{0}$ (positive, so the bracket is positive), and $\mathbf{L}$, $\mathbf{S}$ are in units of $\hbar$: $\mathbf{L}\cdot\mathbf{S}=l/2$ for $j=l+\frac{1}{2}$ and $-(l+1)/2$ for $j=l-\frac{1}{2}$. In atomic units the prefactor is $\alpha^{2}/2$ hartree. `ANT.Gaussian` gives ξ_{ij} in eV and integrates with a 730-point Newton–Cotes rule between 10^{-12} and 10^{-9} m (0.019–18.9 bohr), within 0.4 % of the exact integral for the basis sets of this database; the factors below were fitted with the same quadrature. $R_{\mathrm{i}}\left(r\right)$ are the radial contracted gaussian-type orbitals (CGTOs), with $N_{\mathrm{i}}=\int_{0}^{\infty}R_{\mathrm{i}}^{2}r^{2}\mathrm{d}r$. Only CGTOs on the same atom and of the same shell type ($L=1$, 2 or 3) contribute to the integral because SOC is an intra-atomic phenomenon [1]. However, for CGTO basis sets with pseudopotentials, a single multiplicative correction to ξ_{ij} (SOCFAC_{L} in Eq. (2), one per shell type, keywords `SOC_FAC_P`, `SOC_FAC_D`, `SOC_FAC_F`) is needed in order to account for the correct effective charge in $V_{\mathrm{eff}}\left(r\right)$ due to the lack of nodal structure near the nucleus in pseudopotentials [1]. Here, we make two minor modifications to improve the above implementation. We use the following Yukawa screening potential:

$$V_{\mathrm{eff}}\left(r\right)=\frac{\left(Z-1\right)\exp\left(-\frac{\ln Z}{r_{c}}\,r\right)+1}{r} \tag{3}$$

where r_{c} is a cutoff, of the order of an atomic radius (2–3 a.u.; the calibrated factors below use 2.0 a.u. for all pseudopotential basis sets, keyword `RCUT`) and Z is the atomic number. The cutoff varies between shells, being larger for the inner-most shells and smaller for the outer-most ones. For simplicity, we will choose a common value that suits better the shell or shells dominating at the Fermi level (typically $L=1$, 2 or 3), although multiplicative factors can be used to tune the strength of SOC for the other shells. For instance, for transition metals, the strength for $L=1$ shells is usually 3-5 times that of the $L=2$ shells [9] and this is something that a common cutoff radius may not capture.

Calibration of the SOC factors (2026)

The factors below replace those of the first version of this database. Those were fitted with a script that, when building $H(\mathbf{k})$ from the lower-triangular Fock matrices printed by `CRYSTAL`, counted every SOC matrix element above the diagonal twice (non-Hermitian $H(\mathbf{k})$); `ANT.Gaussian` adds the SOC term correctly. The factors used until now (as shipped with `ANT.UI` 1.0) were therefore not consistent with `ANT.Gaussian`: compared with the present ones, they gave spin–orbit constants about 20 % too weak for Au, and too strong for Pt and Ag ($r_c = 2.5$ bohr; by 20–30 % in the d shells and 40–110 % in the p shells). The present factors are obtained as follows:

(i) For each element, two `CRYSTAL23` [10] calculations of the bulk crystal with the basis set listed below and the Stuttgart small-core pseudopotential (`STUTSC` library): a scalar-relativistic one, and a self-consistent two-component one including the spin–orbit part of the same pseudopotential. Both share geometry, functional (PBE), $\mathbf{k}$ mesh and smearing.

(ii) The SOC term of Eqs. (1)–(3) is computed exactly as in `ANT.Gaussian` (same radial quadrature and units), added to the on-site block of the scalar-relativistic Hamiltonian, and $\mathrm{SOCFAC}_{P,D,F}$ are fitted by least squares to the two-component bands (same basis, so bands correspond one to one) between $E_F - 6$ and $E_F + 4$ eV, allowing a rigid shift. r_c is kept at 2 bohr; letting it free does not improve the fit.

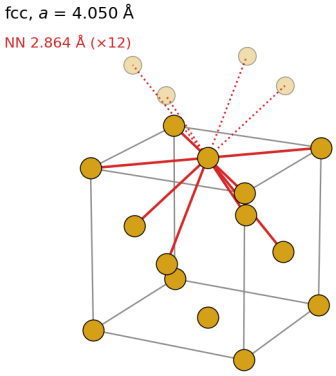
The scalar part of `STUTSC` coincides term by term with the pseudopotentials listed below (checked for each element), so the factors apply to `ANT.Gaussian` runs with exactly these basis sets and pseudopotentials. For all-electron basis sets (Al, Cu) all factors are 1.

Two remarks on the fitted values. (i) The bands in the window hardly constrain SOC_FAC_F: changing it between 0 and several hundred changes the deviation by less than 0.1 meV. It is therefore taken from a least-squares projection of the on-site block of the `CRYSTAL` SOC matrix onto the model, and set to 0 where that projection is negative (`ANT.Gaussian` rejects negative factors). (ii) The fit allows a rigid shift of the bands, between −169 meV (Bi) and +6 meV (Au). It reflects the change of the self-consistent potential when SOC is included and has no consequence in `ANT.Gaussian`, which determines its own Fermi level.

Using the factors in `ANT.Gaussian`

- The factors belong to the basis set *and* the pseudopotential listed for each element, and to the cutoff r_c given with them. With any other basis set, pseudopotential or `RCUT` they are not valid.
- `SOC_FAC_P`, `SOC_FAC_D`, `SOC_FAC_F` and `RCUT` apply to every atom. For several elements, or atoms with a different basis set, use `SOCFACATOM`: the number of entries, then one line `atom, P, D, F, RCUT` per atom. Unlisted atoms keep the global values, so `SOCFACATOM` goes after the global keywords.
- The log lists the resulting spin–orbit constants ξ_{ij} ("Spin-orbit constants", in eV), a quick check that the intended factors were read.

Au (Gold): face-centred cubic



Au at a glance

Structure	face-centred cubic, $Fm\bar{3}m$; $a = 4.05$ Å
Basis set	pob-TZVP-rev2 + $s(0.1)$ [11]; (7s6p6d1f)/[5s3p3d1f], 36 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 60 core / 19 valence electrons; h local; s–h (2), p–h (4), d–h (4), f–h (2), g–h (2)
SOC factors	P 201.3, D 43.2, F 12.6; $r_c = 2.0$ bohr
Fit quality	RMS 10.3 meV, max 48.9 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All Au atoms with this basis set:

```
SOC
SOC_FAC_P = 201.3
SOC_FAC_D = 43.2
SOC_FAC_F = 12.6
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 201.3, 43.2, 12.6, 2.0
2, 201.3, 43.2, 12.6, 2.0
```

Calibration

Basis set: the `Au_pob_TZVP_rev2` basis set plus an additional uncontracted s primitive (exponent 0.1) [11], with the Stuttgart small-core pseudopotential. Structure: a face-centred cubic (FCC) lattice with $a = 4.05$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 10.3 meV, maximum 48.9 meV, rigid shift 6 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site `CRYSTAL` SOC matrix onto the model.

On-site projection, for comparison: SOC_FAC_P = 231.2, SOC_FAC_D = 43.7, SOC_FAC_F = 12.6.

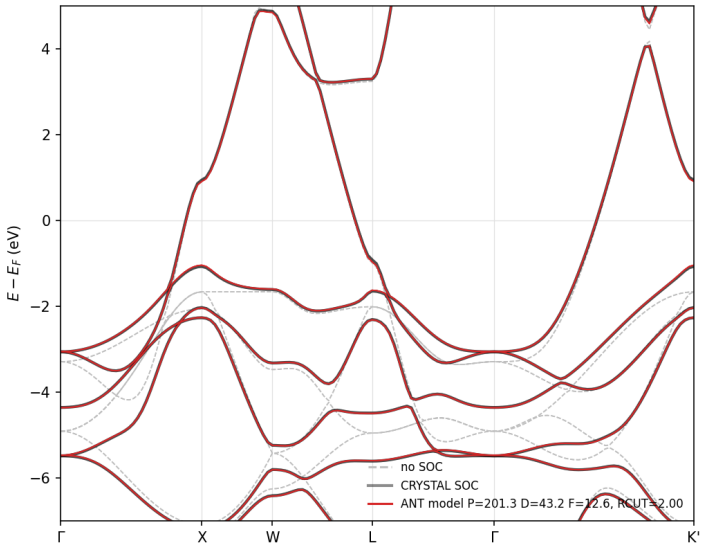


Figure 1. Au: bands of the self-consistent two-component CRYSTAL23 calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

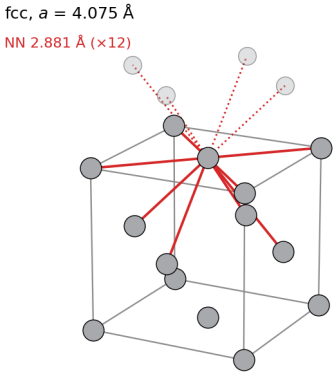
Basis set: pob-TZVP-rev2 + s(0.1), (7s6p6d1f)/[5s3p3d1f] **32 lines**

```
Au 0
S 3 1.0
30.0000000000 0.207492311080
27.0000000000 -0.332678933940
14.7468243310 0.383028179580
S 1 1.0
6.10568239000 1.000000000000
S 1 1.0
1.13068946000 1.000000000000
S 1 1.0
0.18001283000 1.000000000000
S 1 1.0
0.10000000000 1.000000000000
P 4 1.0
15.5000000000 0.150017118800
14.0000000000 -0.236098131830
6.42273682050 0.314588969480
1.65956016810 -0.572796704460
P 1 1.0
0.87155106000 1.000000000000
P 1 1.0
0.20887895000 1.000000000000
D 4 1.0
9.55240986560 0.040145559502
7.26988869370 -0.093690906606
1.77464967890 0.317462823170
0.79960541055 0.467951924830
D 1 1.0
0.55708019000 1.000000000000
D 1 1.0
0.28104926000 1.000000000000
F 1 1.0
0.72482000000 1.000000000000
```

Pseudopotential: 60 core electrons, scalar-relativistic part

```
Au 0
AU-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 13.52321800 426.64186700
2 6.26438400 36.80066800
p-h potential
4
2 11.41386700 87.00209100
2 10.32921500 174.00437000
2 5.70742400 8.87061000
2 4.82816500 17.90243800
d-h potential
4
2 7.43096300 49.88365500
2 8.32199000 74.68454900
2 4.60964200 6.48622700
2 3.51150700 9.54682100
f-h potential
2
2 3.08463900 8.79164000
2 3.02474300 11.65845600
g-h potential
2
2 3.97844200 -5.23433700
2 4.01149100 -6.73814200
```

Ag (Silver): face-centred cubic



Ag at a glance

Structure	face-centred cubic, $Fm\bar{3}m$; $a = 4.075$ Å
Basis set	pob-DZVP-2018 [12]; (5s5p5d1f)/[3s2p2d1f], 26 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 28 core / 19 valence electrons; g local; s–g (2), p–g (4), d–g (4), f–g (2)
SOC factors	P 49.1, D 8.9, F 0.1; $r_c = 2.0$ bohr
Fit quality	RMS 3.4 meV, max 18.2 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All Ag atoms with this basis set:

```
SOC
SOC_FAC_P = 49.1
SOC_FAC_D = 8.9
SOC_FAC_F = 0.1
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 49.1, 8.9, 0.1, 2.0
2, 49.1, 8.9, 0.1, 2.0
```

Calibration

Basis set: the `Ag_pob_DZVP_2018` basis set [12], with the Stuttgart small-core pseudopotential. Structure: an FCC lattice with $a = 4.075$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 3.4 meV, maximum 18.2 meV, rigid shift -2 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site `CRYSTAL` SOC matrix onto the model. The bands depend only weakly on SOC_FAC_P (RMS 3.4–4.1 meV for SOC_FAC_P between 0 and 85): the SOC splittings in the window come from the d shells. The value given is the least-squares optimum. On-site projection, for comparison: SOC_FAC_P = 84.7, SOC_FAC_D = 8.2, SOC_FAC_F = 0.1.

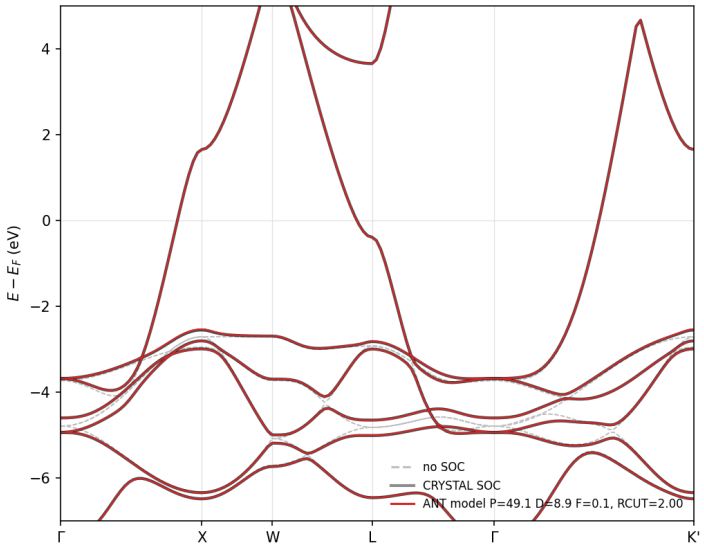


Figure 2. Ag: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

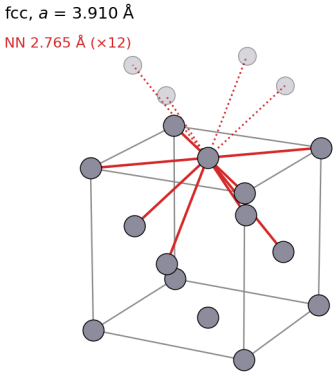
Basis set: pob-DZVP-2018, (5s5p5d1f)/[3s2p2d1f] — **24 lines**

```
Ag 0
S 3 1.0
9.08844200000 -1.980891879700
7.54073100000 2.755451334700
2.79400500000 0.227154083810
S 1 1.0
1.23043317000 1.000000000000
S 1 1.0
0.18387114000 1.000000000000
P 4 1.0
4.45124000000 -0.993521037710
3.67526300000 1.050052523700
1.26106209050 0.647475325370
0.54212477498 0.256215507230
P 1 1.0
0.15011983000 1.000000000000
D 4 1.0
7.79566722920 -0.017042912377
2.89265102380 0.234461548030
1.24742732030 0.447658775330
0.49313817671 0.390649545600
D 1 1.0
0.15938987000 1.000000000000
F 1 1.0
1.39711000000 1.000000000000
```

Pseudopotential: 28 core electrons, scalar-relativistic part

```
Ag 0
AG-ECP 4 28
g potential
1
2 1.0000000 0.0000000
s-g potential
2
2 12.5677140 255.0547710
2 6.9976620 36.9833930
p-g potential
4
2 11.3164960 60.7157050
2 10.9580630 121.4438890
2 7.1114000 10.1718660
2 6.7733190 20.4865640
d-g potential
4
2 8.9284370 29.5049380
2 11.1025670 44.0187360
2 5.5432120 5.3683330
2 3.9288350 7.4083750
f-g potential
2
2 11.0129130 -12.6234030
2 11.0198980 -16.7643270
```

Pt (Platinum): face-centred cubic



Pt at a glance

Structure	face-centred cubic, $Fm\bar{3}m$; $a = 3.91$ Å
Basis set	pob-TZVP-rev2 [11]; (6s6p6d1f)/[4s3p3d1f], 35 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 60 core / 18 valence electrons; h local; s–h (2), p–h (4), d–h (4), f–h (2), g–h (2)
SOC factors	P 229.4, D 41.6, F 9.3; $r_c = 2.0$ bohr
Fit quality	RMS 5.0 meV, max 24.1 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All Pt atoms with this basis set:

```
SOC
SOC_FAC_P = 229.4
SOC_FAC_D = 41.6
SOC_FAC_F = 9.3
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 229.4, 41.6, 9.3, 2.0
2, 229.4, 41.6, 9.3, 2.0
```

Calibration

Basis set: the `Pt_pob_TZVP_rev2` basis set [11], with the Stuttgart small-core pseudopotential. Structure: an FCC lattice with $a = 3.91$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 5.0 meV, maximum 24.1 meV, rigid shift -23 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site `CRYSTAL` SOC matrix onto the model.

On-site projection, for comparison: SOC_FAC_P = 237.3, SOC_FAC_D = 42.6, SOC_FAC_F = 9.3.

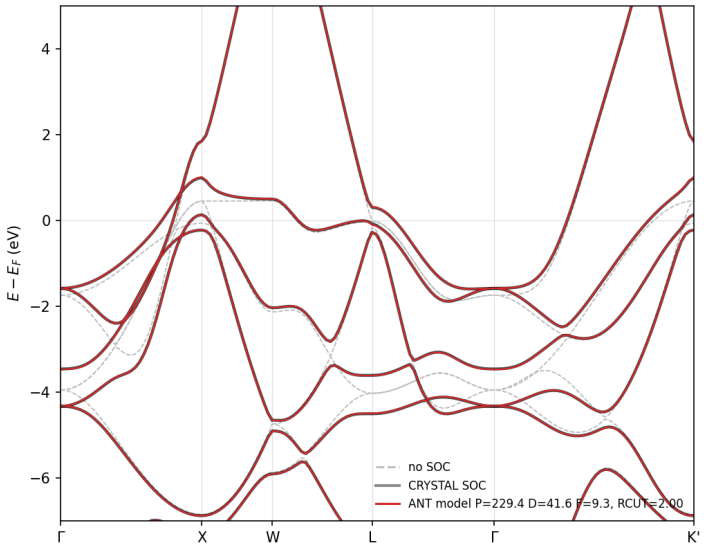


Figure 3. Pt: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

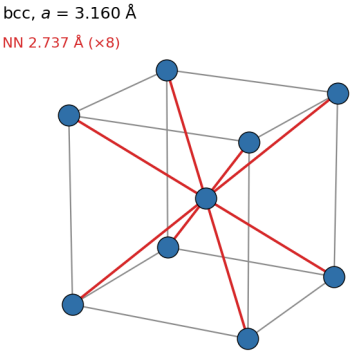
Basis set: pob-TZVP-rev2, (6s6p6d1f)/[4s3p3d1f] — **30 lines**

```
Pt 0
S 3 1.0
  30.0000000000 0.271482639000
  27.0000000000 -0.422267587740
  14.4083185640 0.443615819950
S 1 1.0
  5.76260815000 1.000000000000
S 1 1.0
  1.06835871000 1.000000000000
S 1 1.0
  0.15368971000 1.000000000000
P 4 1.0
  15.5000000000 -0.156727186290
  14.0000000000 0.238534129890
  6.11612123390 -0.310413797330
  1.57155863850 0.564735250890
P 1 1.0
  0.77232013000 1.000000000000
P 1 1.0
  0.25285881000 1.000000000000
D 4 1.0
  8.32079376110 0.062945798646
  7.42072265200 -0.090271847072
  1.65704106390 0.168125264160
  0.73943569960 0.250454169700
D 1 1.0
  1.12524984000 1.000000000000
D 1 1.0
  0.27562365000 1.000000000000
F 1 1.0
  0.56813000000 1.000000000000
```

Pseudopotential: 60 core electrons, scalar-relativistic part

```
Pt 0
PT-ECP 5 60
h potential
1
2 1.00000000 0.00000000
s-h potential
2
2 14.60450000 429.64608700
2 7.21828700 73.15688400
p-h potential
4
2 11.57716200 88.02291700
2 10.88384300 175.99819600
2 6.42440300 13.68227400
2 5.22419800 27.41465100
d-h potential
4
2 7.69961000 43.55785200
2 7.55080800 65.36910800
2 3.96116400 7.01859600
2 3.87277700 11.39173300
f-h potential
2
2 3.37986900 10.71022000
2 3.32625500 14.27812500
g-h potential
2
2 5.45202000 -11.65174900
2 5.41258500 -14.37552500
```

W (Tungsten): body-centred cubic



W at a glance

Structure	body-centred cubic, $Im\bar{3}m$; $a = 3.16$ Å
Basis set	pob-TZVP-rev2 (mod.) [11]; (7s6p6d1f)/[5s3p3d1f], 36 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 60 core / 14 valence electrons; h local; s–h (2), p–h (6), d–h (6), f–h (2), g–h (2)
SOC factors	P 241.0, D 55.4, F 10.7; $r_c = 2.0$ bohr
Fit quality	RMS 5.8 meV, max 27.7 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All W atoms with this basis set:

```
SOC
SOC_FAC_P = 241.0
SOC_FAC_D = 55.4
SOC_FAC_F = 10.7
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 241.0, 55.4, 10.7, 2.0
2, 241.0, 55.4, 10.7, 2.0
```

Calibration

Basis set: the `W_pob_TZVP_rev2` basis set with additional diffuse s functions and re-optimized outer exponents [11], with the Stuttgart small-core pseudopotential. Structure: a body-centred cubic (BCC) lattice with $a = 3.16$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 5.8 meV, maximum 27.7 meV, rigid shift -72 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is therefore taken from the projection of the on-site `CRYSTAL` SOC matrix onto the model.

On-site projection, for comparison: SOC_FAC_P = 248.3, SOC_FAC_D = 55.1, SOC_FAC_F = 10.7.

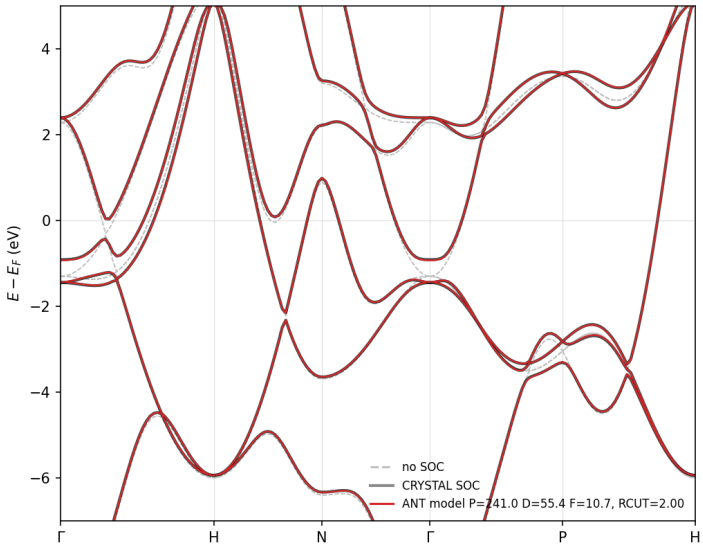


Figure 4. W: bands of the self-consistent two-component `CRYSTAL23` calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

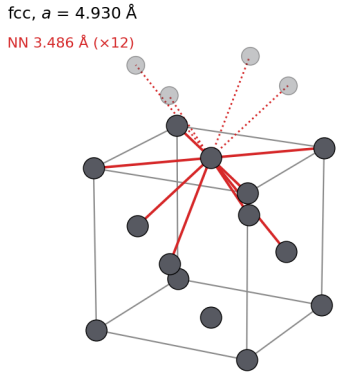
Basis set: pob-TZVP-rev2 (mod.), (7s6p6d1f)/[5s3p3d1f] **32 lines**

```
W 0                              S 1 1.0                          D 4 1.0
S 3 1.0                          0.117 1.000000000000             7.40647373150 0.086993963018
30.0000000000 0.322464834100     P 4 1.0                          5.90262686030 -0.176675400110
27.0000000000 -0.466922572140    17.3684 -0.0364                  1.29847567500 0.551456970300
13.0780456840 0.426995637760     12.4136 0.1092                   0.57153508541 0.953135965350
S 1 1.0                          5.15862176580 -0.293999550200    D 1 1.0
4.56489858000 1.000000000000     1.2804 0.5082                    0.4269 1.000000000000
S 1 1.0                          P 1 1.0                          D 1 1.0
0.92909758000 1.000000000000     0.6577 1.000000000000            0.2466 1.000000000000
S 1 1.0                          P 1 1.0                          F 1 1.0
0.2029 1.000000000000            0.245 1.000000000000             0.3026 1.000000000000
```

Pseudopotential: 60 core electrons, scalar-relativistic part

```
W 0                              2 9.3381880 107.3481100          4 6.2263750 0.0219560
W-ECP 5 60                       2 8.4304480 214.6995680          2 1.9638750 -0.0885770
h potential                      4 9.4900200 0.0254420            2 1.8882870 -0.2097260
1                                4 9.4899470 0.0518950            f-h potential
2 1.00000000 0.00000000          2 1.8829970 -0.1171840           2
s-h potential                    2 1.9069720 0.2966890            2 2.3079530 6.2324720
2                                d-h potential                    2 2.2706090 8.3113450
2 11.0637950 419.2275990         6                                g-h potential
2 8.2176410 41.1913070           2 6.2054330 58.8812790           2
p-h potential                    2 6.1221570 98.6835560           2 3.5834910 -6.8029440
6                                4 6.2745560 0.0195370            2 3.5625150 -8.4432320
```

Pb (Lead): face-centred cubic



Pb at a glance

Structure	face-centred cubic, $Fm\bar{3}m$; $a = 4.93$ Å
Basis set	Baranek 2013 [13]; (9s9p6d1f)/[4s4p2d1f], 33 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 60 core / 22 valence electrons; h local; s–h (2), p–h (4), d–h (4), f–h (2), g–h (2)
SOC factors	P 248.3, D 66.5, F 0.0; $r_c = 2.0$ bohr
Fit quality	RMS 9.9 meV, max 51.9 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All Pb atoms with this basis set:

```
SOC
SOC_FAC_P = 248.3
SOC_FAC_D = 66.5
SOC_FAC_F = 0.0
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 248.3, 66.5, 0.0, 2.0
2, 248.3, 66.5, 0.0, 2.0
```

Calibration

Basis set: the optimized basis set of Baranek and co-workers [13], with the Stuttgart small-core pseudopotential. Structure: an FCC lattice with $a = 4.93$ Å. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 9.9 meV, maximum 51.9 meV, rigid shift -124 meV, with r_c fixed at 2.0 bohr. The bands are insensitive to SOC_FAC_F, which is fixed at 0. The pseudopotential is the complete Stuttgart small-core one (60 core electrons) including its g projector; the 2021 `CRYSTAL` input of this basis set omitted the two g terms. The projection gives a negative SOC_FAC_F, which ANT does not accept, hence 0. SOC_FAC_D is fixed by the bands even though the $5d$ shell lies below the window (RMS 18 meV for SOC_FAC_D = 0, 9.9 meV at the optimum). On-site projection, for comparison: SOC_FAC_P = 207.3, SOC_FAC_D = 38.1, SOC_FAC_F = −32.8.

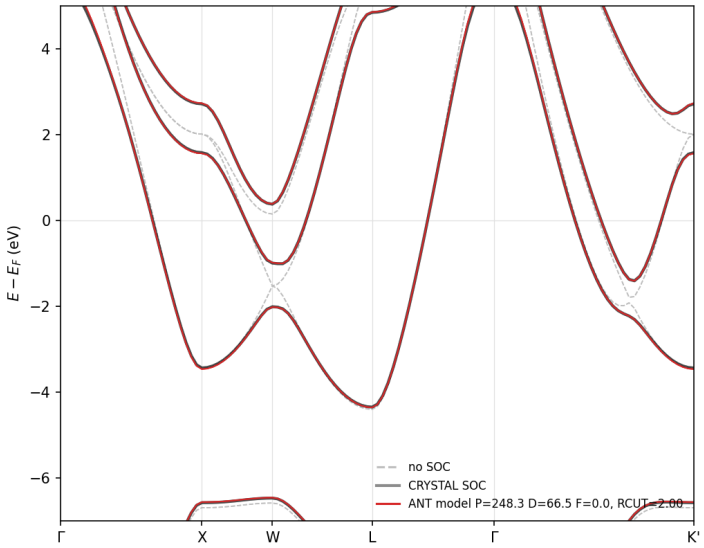


Figure 5. Pb: bands of the self-consistent two-component CRYSTAL23 calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

Basis set: Baranek 2013, (9s9p6d1f)/[4s4p2d1f] — **36 lines**

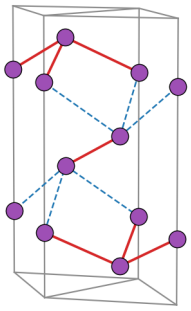
```
Pb 0                                0.147 1.                        0.1325 1.
S 6 1.0                             P 6 1.0                         D 5 1.0
8.55351735988 3.88475299531         8.55351735988 3.13503035185     11.1394028684 0.299749731795
7.35277444701 -5.11944720447        7.35277444701 -4.65435922478    7.57317950313 -0.908307739883
2.95283552202 0.359342355194        2.95283552202 1.34852524939     2.21243848295 3.44990840575
1.43101525276 2.0148141461          1.43101525276 3.23612718594     1.0927431021 5.08865830004
0.659101026799 0.477751455641       0.659101026799 1.35196984634    0.517878712738 3.46485836421
0.220157140772 -0.029791382408      0.220157140772 0.0649906557872  D 1 1.0
S 1 1.0                             P 1 1.0                         0.2314 1.
1.2543 1.                           1.2879 1.                       F 1 1.0
S 1 1.0                             P 1 1.0                         0.1354 1.0
0.4504 1.                           0.5176 1.
S 1 1.0                             P 1 1.0
```

Pseudopotential: 60 core electrons, scalar-relativistic part

```
Pb 0                                4                               2 4.5637890 14.8337310
PB-ECP 5 60                         2 10.2417900 72.2768970         f-h potential
h potential                         2 8.9241760 144.5910830         2
1                                   2 6.5813420 4.7586930           2 3.8875120 12.2098920
2 1.00000000 0.00000000             2 6.2554030 9.9406230           2 3.8119630 16.1902910
s-h potential                       d-h potential                   g-h potential
2                                   4                               2
2 12.2963030 281.2854990            2 7.7543360 35.8485070          2 5.6915770 -9.0966650
2 8.6326340 62.5202170              2 7.7202810 53.7243420          2 5.7155670 -11.5319960
p-h potential                       2 4.9702640 10.1152560
```

Bi (Bismuth): rhombohedral A7

A7 ($R\bar{3}m$), a = 4.546 Å, c = 11.862 Å, u = 0.2339
short 3.071 Å (×3, within a bilayer)
long 3.529 Å (×3, between bilayers)



Bi at a glance

Structure	rhombohedral A7, $R\bar{3}m$; $a = 4.546$, $c = 11.862$ Å, $u = 0.2339$ (hexagonal axes, 2 atoms per primitive cell)
Basis set	pob-TZVP-rev2 + $s(0.1)$ [11]; (10s8p9d)/[5s4p4d], 37 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 60 core / 23 valence electrons; h local; s–h (2), p–h (4), d–h (4), f–h (2), g–h (2)
SOC factors	P 230.9, D 24.1, F 0.0; $r_c = 2.0$ bohr
Fit quality	RMS 11.7 meV, max 41.1 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All Bi atoms with this basis set:

```
SOC
SOC_FAC_P = 230.9
SOC_FAC_D = 24.1
SOC_FAC_F = 0.0
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 230.9, 24.1, 0.0, 2.0
2, 230.9, 24.1, 0.0, 2.0
```

Calibration

Basis set: the `Bi_pob_TZVP_rev2` basis set plus an additional uncontracted s primitive (exponent 0.1) [11], with the Stuttgart small-core pseudopotential. Structure: the rhombohedral A7 structure ($R\bar{3}m$) with the experimental room-temperature hexagonal parameters $a = 4.546$ Å, $c = 11.862$ Å and $u = 0.2339$. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 11.7 meV, maximum 41.1 meV, rigid shift -169 meV, with r_c fixed at 2.0 bohr. The basis set has no f shells. SOC_FAC_P is well defined (229–233 for any SOC_FAC_D between 0 and 60), SOC_FAC_D only weakly (RMS 11.7–12.7 meV over that range). The on-site projection of SOC_FAC_P is dominated by the semicore $5p$ shell and is not representative of the valence $6p$ bands. On-site projection, for comparison: SOC_FAC_P = 114.9, SOC_FAC_D = 34.7.

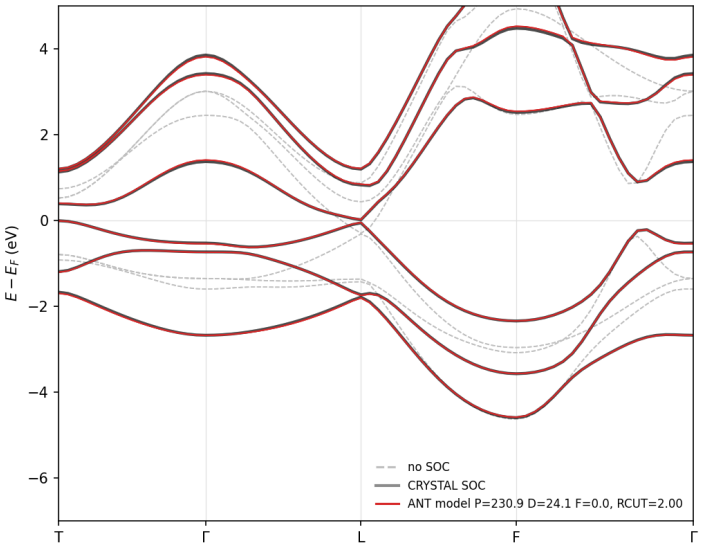


Figure 6. Bi: bands of the self-consistent two-component CRYSTAL23 calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

Basis set: pob-TZVP-rev2 + s(0.1), (10s8p9d)/[5s4p4d] — **40 lines**

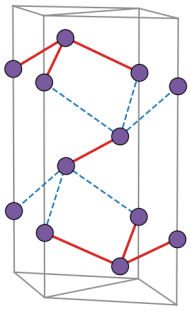
```
Bi 0                                    S 1 1.0                                 D 6 1.0
S 4 1.0                                 0.10000000000 1.000000000000            66.4044819480 0.000381028783
716.414353100 0.000312543071           P 3 1.0                                 13.8584269610 0.010746152442
83.8060590470 0.001762476895            15.2496446690 0.745603560000            7.06545190000 -0.071947646845
21.1169628530 -0.219109834370           14.8461760530 -0.855786373380           2.52521440350 0.261959749890
15.4914481870 0.404112249310            7.06368267840 0.401491595920            1.34195850000 0.425947500000
S 2 1.0                                 P 3 1.0                                 0.68340941000 0.336803256270
23.2398550290 -0.068255758685           2.58812556160 0.355427296330            D 1 1.0
6.64742550000 0.978880464710            1.50202084990 0.639769918900            0.90438000000 1.000000000000
S 2 1.0                                 0.76732724388 0.323327738390            D 1 1.0
1.96177440000 0.653863290000            P 1 1.0                                 0.41457000000 1.000000000000
0.97252866000 0.346136700000            0.42288904000 1.000000000000            D 1 1.0
S 1 1.0                                 P 1 1.0                                 0.15146817000 1.000000000000
0.20618896000 1.000000000000            0.17062087000 1.000000000000
```

Pseudopotential: 60 core electrons, scalar-relativistic part

```
Bi 0                                    4                                       2 4.7045590 14.9814850
BI-ECP 5 60                             2 10.4677770 72.0014990                 f-h potential
h potential                             2 9.1189010 144.0022770                 2
1                                       2 6.7547910 5.0079450                   2 4.2145460 13.7133830
2 1.00000000 0.00000000                 2 6.2525920 9.9915500                   2 4.1334000 18.1943080
s-h potential                           d-h potential                           g-h potential
2                                       4                                       2
2 13.0430900 283.2642270                2 8.0814740 36.3962590                  2 6.2057090 -10.2474430
2 8.2216820 62.4719590                  2 7.8905950 54.5976640                  2 6.2277820 -12.9557100
p-h potential                           2 4.9555560 9.9842940
```

Sb (Antimony): rhombohedral A7

A7 ($R\bar{3}m$), a = 4.3007 Å, c = 11.222 Å, u = 0.2335
short 2.901 Å (×3, within a bilayer)
long 3.345 Å (×3, between bilayers)



Sb at a glance

Structure	rhombohedral A7, $R\bar{3}m$; $a = 4.3007$, $c = 11.222$ Å, $u = 0.2335$ (hexagonal axes, 2 atoms per primitive cell)
Basis set	pob-TZVP-2018 [12]; (8s8p9d)/[4s4p4d], 36 functions per atom (spherical d, f)
Pseudo-potential	Stuttgart small-core, 28 core / 23 valence electrons; g local; s–g (2), p–g (4), d–g (4), f–g (2)
SOC factors	P 76.9, D 0.0, F 0.0; $r_c = 2.0$ bohr
Fit quality	RMS 5.8 meV, max 24.4 meV ($E_F - 6$ to $E_F + 4$ eV)

ANT.Gaussian keywords (`.ant` file)

All Sb atoms with this basis set:

```
SOC
SOC_FAC_P = 76.9
SOC_FAC_D = 0.0
SOC_FAC_F = 0.0
RCUT      = 2.0
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 76.9, 0.0, 0.0, 2.0
2, 76.9, 0.0, 0.0, 2.0
```

Calibration

Basis set: the `Sb_pob_TZVP_2018` basis set [12], with the Stuttgart small-core pseudopotential. Structure: the rhombohedral A7 structure ($R\bar{3}m$) with the experimental room-temperature hexagonal parameters $a = 4.3007$ Å, $c = 11.222$ Å and $u = 0.2335$. Reference: self-consistent two-component `CRYSTAL23` [10] calculation (PBE, $32 \times 32 \times 32$ Monkhorst–Pack mesh, Fermi smearing 0.001 Ha) with the same basis set and the spin–orbit part of the same pseudopotential (`STUTSC` library). The factors are fitted to its bands between $E_F - 6$ and $E_F + 4$ eV (path plus a $6 \times 6 \times 6$ mesh): RMS deviation 5.8 meV, maximum 24.4 meV, rigid shift -12 meV, with r_c fixed at 2.0 bohr. The basis set has no f shells. The bands prefer no SOC on the d shells: RMS 5.8 meV for SOC_FAC_D = 0, 6.3 meV at the on-site projection value 8.1 and 10.2 meV at 40 (SOC_FAC_P stays between 77 and 81). SOC_FAC_D ≈ 8 is an acceptable alternative if the semicore $4d$ splitting matters. On-site projection, for comparison: SOC_FAC_P = 38.9, SOC_FAC_D = 8.1.

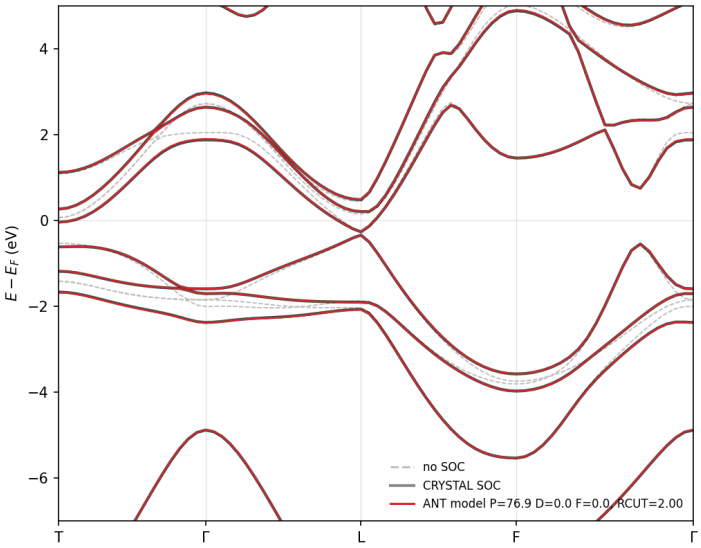


Figure 7. Sb: bands of the self-consistent two-component CRYSTAL23 calculation (grey), of the scalar-relativistic Hamiltonian plus the ANT on-site SOC with the factors above (red), and without SOC (dashed).

Basis set and pseudopotential (Gaussian format)

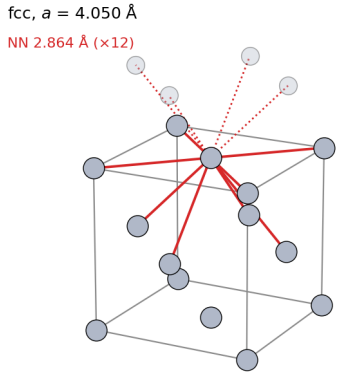
Basis set: pob-TZVP-2018, (8s8p9d)/[4s4p4d] **37 lines**

```
Sb 0
S 4 1.0
1612.41999330 0.000285403808
238.844520970 0.001339377875
23.9981188090 -0.049388154574
15.1931242130 0.433922272540
S 2 1.0
11.7364097330 0.921255199650
6.52597747940 0.792352802260
S 1 1.0
1.58348739000 1.000000000000
S 1 1.0
0.18345132000 1.000000000000
P 3 1.0
215.683933540 0.000260518232
16.3744790880 0.073728000195
9.72162833450 -0.272300281280
P 3 1.0
2.79826431540 0.464726923740
1.47110450330 0.503642420750
0.75165385301 0.187066662940
P 1 1.0
0.37211239000 1.000000000000
P 1 1.0
0.12000001000 1.000000000000
D 6 1.0
115.903122530 0.000531409151
30.4742337200 0.005941113917
18.2284182390 -0.010563706947
4.32914566460 0.203481773410
2.12948184960 0.427483789280
0.99682636692 0.385395608090
D 1 1.0
0.62366245000 1.000000000000
D 1 1.0
0.31224235000 1.000000000000
D 1 1.0
0.12000012000 1.000000000000
```

Pseudopotential: 28 core electrons, scalar-relativistic part

```
Sb 0
SB-ECP 4 28
g potential
1
2 1.00000000 0.00000000
s-g potential
2
2 16.3308650 281.0715810
2 8.5565420 61.7166040
p-g potential
4
2 14.4703370 67.4573800
2 13.8161940 134.9335030
2 8.4249240 14.7163440
2 8.0927280 29.5185120
d-g potential
4
2 14.8863310 35.4478150
2 15.1463190 53.1434660
2 5.9082670 9.1792230
2 5.5943220 13.2402530
f-g potential
2
2 14.4449780 -15.3668010
2 14.4492950 -20.2961380
```

Al (Aluminium): face-centred cubic



Al at a glance

Structure	face-centred cubic, $Fm\bar{3}m$; $a = 4.05$ Å
Basis set	pob-DZVP-rev2 [14]; (10s7p1d)/[4s3p1d], 18 functions per atom (spherical d, f)
Pseudo-potential	none (all-electron)
SOC factors	P 1.0, D 1.0, F 0.0; $r_c = 2.5$ bohr
Fit quality	not recalibrated (all-electron basis set)

ANT.Gaussian keywords (`.ant` file)

All Al atoms with this basis set:

```
SOC
SOC_FAC_P = 1.0
SOC_FAC_D = 1.0
SOC_FAC_F = 0.0
RCUT      = 2.5
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 1.0, 1.0, 0.0, 2.5
2, 1.0, 1.0, 0.0, 2.5
```

SOC factors

Basis set: the all-electron `Al_pob_DZVP_rev2` basis set [14]. All-electron basis sets keep the nodal structure of the radial functions near the nucleus, which Eq. (2) assumes, so no multiplicative correction is applied (the 2021 fits against `OpenMX` also ended at factors of 1). Not recalibrated: `CRYSTAL23` includes SOC only through pseudopotentials with spin–orbit terms, so there is no same-basis reference for all-electron basis sets.

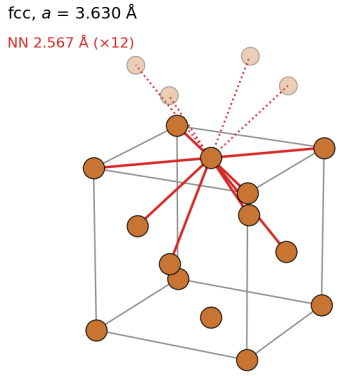
Basis set and pseudopotential (Gaussian format)

Basis set: pob-DZVP-rev2, (10s7p1d)/[4s3p1d] — 26 lines

```
Al 0
S 5 1.0
5887.5727030 0.0013483347987
885.61225996 0.0100715768090
201.13604899 0.0451324540560
56.284974674 0.1146126804300
17.229551243 0.1015960894300
S 3 1.0
29.340249922 0.0693474542080
3.0439630420 -0.4252811767900
1.1285539518 -0.4144983221000
S 1 1.0
0.7834298000 1.0000000000000
S 1 1.0
0.1440019200 1.0000000000000
P 5 1.0
145.11918809 0.0063963373134
33.717894833 0.0441893599650
10.369863083 0.1558157599300
3.5135616036 0.2863528695100
1.1980050273 0.2292142324800
P 1 1.0
0.3200672400 1.0000000000000
P 1 1.0
0.1500336200 1.0000000000000
D 1 1.0
0.1530089300 1.0000000000000
```

No pseudopotential: all-electron basis set.

Cu (Copper): face-centred cubic



Cu at a glance

Structure	face-centred cubic, $Fm\bar{3}m$; $a = 3.63$ Å
Basis set	Ruiz 2003 [15]; (14s10p5d)/[6s6p3d], 39 functions per atom (spherical d, f)
Pseudo-potential	none (all-electron)
SOC factors	P 1.0, D 1.0, F 0.0; $r_c = 2.5$ bohr
Fit quality	not recalibrated (all-electron basis set)

ANT.Gaussian keywords (`.ant` file)

All Cu atoms with this basis set:

```
SOC
SOC_FAC_P = 1.0
SOC_FAC_D = 1.0
SOC_FAC_F = 0.0
RCUT      = 2.5
```

Per atom (here atoms 1 and 2), after the global keywords:

```
SOCFACATOM
2
1, 1.0, 1.0, 0.0, 2.5
2, 1.0, 1.0, 0.0, 2.5
```

SOC factors

Basis set: the all-electron basis set of Ruiz *et al.* [15]. All-electron basis sets keep the nodal structure of the radial functions near the nucleus, which Eq. (2) assumes, so no multiplicative correction is applied (the 2021 fits against `OpenMX` also ended at factors of 1). Not recalibrated: `CRYSTAL23` includes SOC only through pseudopotentials with spin–orbit terms, so there is no same-basis reference for all-electron basis sets.

Basis set and pseudopotential (Gaussian format)

Basis set: Ruiz 2003, (14s10p5d)/[6s6p3d] — 44 lines

```
Cu 0
S 6 1.0
76441.483694 0.0014236548869
11477.790484 0.010910644019
2611.4773990 0.054139319495
737.33531101 0.18863206957
240.02819227 0.38341646126
82.721112667 0.29645681555
S 3 1.0
160.07447183 -0.11037499013
18.854087693 0.64586203973
7.7438526664 0.44511050812
S 2 1.0
13.678658833 -0.22684755916
2.2583597115 0.72397240174
S 1 1.0
.92525661579 1.0000000000
S 1 1.0
.41000000000 1.0000000000
S 1 1.0
.18000000000 1.0000000000
P 3 1.0
2530.0174404 0.0019142187743
600.09192696 0.015800163021
194.09448024 0.076268471159
P 3 1.0
73.686313161 0.23880092118
30.458500032 0.44974987602
13.130879763 0.39361952224
P 1 1.0
5.5263086949 1.0000000000
P 1 1.0
2.1487388219 1.0000000000
P 1 1.0
0.8145 1.0000000000
P 1 1.0
0.1327 1.0000000000
D 3 1.0
51.443304983 0.029319456167
14.404225180 0.15699236141
4.8477556014 0.37790966643
D 1 1.0
1.6194940205 1.0000000000
D 1 1.0
0.4664 1.0000000000
```

No pseudopotential: all-electron basis set.

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References

[1] Pakdel S, Pourfath M and Palacios J J 2018 *Beilstein J. Nanotechnol.* **9** 1015

[2] Kresse G and Furthmüller J 1996 *Computational Materials Science* **6** 15–50

[3] Blaha P, Schwarz K, Madsen G, Kvasnicka D and Luitz J 2001 *Wien2k: An augmented plane wave+ local orbitals program for calculating crystal properties* (Karlheinz Schwarz, Techn. Universität Wien, Austria) ISBN 3-9501031-1-2

[4] Giannozzi P *et al.* 2009 *J. Phys.: Condens. Matter* **21** 395502

[5] Ozaki T 2003 *Phys. Rev. B* **67**(15) 155108

[6] Ozaki T and Kino H 2004 *Phys. Rev. B* **69**(19) 195113

[7] Ozaki T and Kino H 2005 *Phys. Rev. B* **72**(4) 045121

[8] Ozaki T *et al.* 2017 OpenMX (Open source package for Material eXplorer) ver. 3.8 URL http://www.openmx-square.org/

[9] Barreteau C, Spanjaard D and Desjonquères M C 2016 *Comptes Rendus Physique* **17** 406

[10] Erba A, Desmarais J K, Casassa S, Civalleri B, Donà L, Bush I J, Searle B, Maschio L, Edith-Daga L, Cossard A, Ribaldone C, Ascrizzi E, Marana N L, Flament J P and Kirtman B 2023 *Journal of Chemical Theory and Computation* **19** 6891–6932

[11] Laun J and Bredow T 2021 *Journal of Computational Chemistry* **42** 1064–1072

[12] Laun J, Vilela Oliveira D and Bredow T 2018 *Journal of Computational Chemistry* **39** 1285–1290

[13] Sophia G, Baranek P, Sarrazin C, Rérat M and Dovesi R 2013 *Phase Transitions* **86** 1069–1084

[14] Vilela Oliveira D, Laun J, Peintinger M F and Bredow T 2019 *Journal of Computational Chemistry* **40** 2364–2376

[15] Ruiz E, Llunell M and Alemany P 2003 *Journal of Solid State Chemistry* **176** 400–411